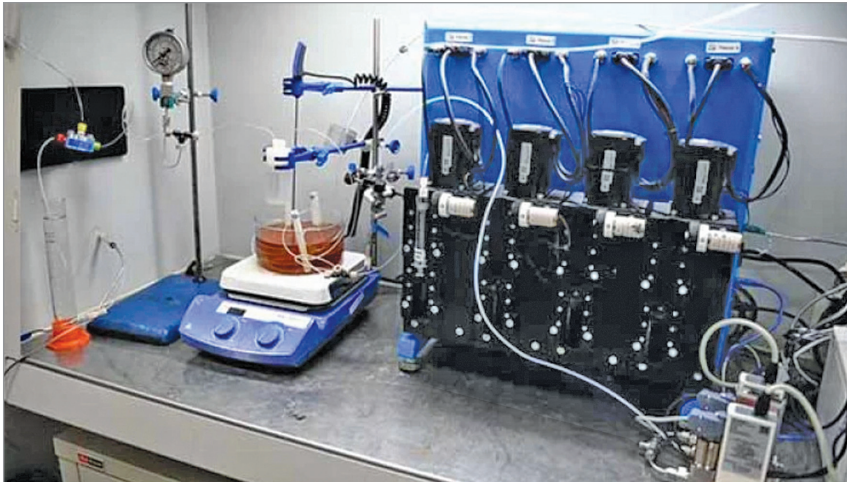




Prototype of an autonomous device capable of extracting water from atmospheric air
(Credit: <https://www.ismatimes.com/en>)

challenge lies in releasing and converting the absorbed moisture into liquid water. This conversion process requires heat energy, typically around 65 to 70 degrees Celsius. This heat energy can be sourced from renewable energy like solar heat, industrial waste heat, biomass waste, or by converting electricity into heat through a heat pump.

Compared to conventional air-conditioner-like systems, which chill surfaces to condense moisture and consume significant power, desiccant-based technology is more efficient. Traditional systems also heavily depend on local humidity levels. In contrast, Uravu Labs' technology adapts its desiccant usage based on the geographic location, ensuring consistent energy consumption and water production across diverse regions.

Advanced filtration processes ensure purity in air-to-water technology

Contamination is a significant issue with water, requiring advanced filtration processes to ensure purity. Ramesha Jha, CEO and Co-founder of Vayujal, explains what sets their technology apart: "We have achieved the remarkable feat of eliminating 90% of particulate

matter from the fresh air we draw from the environment, ensuring the purity and complete absence of contaminants in the water we produce."

Their process begins with drawing in fresh air, which is subjected to advanced filtration that removes a sizeable portion of particulate matter. This meticulous filtration process ensures that the water produced is pure and contaminant-free. Once filtered, the air passes over cold coils set at dewpoint temperatures, causing water vapour to condense and collect in a tray, guided by gravity.

The result is pure, pristine water free from impurities. Additionally, Vayujal's technology can generate 100% renewable water by harnessing energy from sources like solar panels, making them eco-friendly and sustainable. Compared to competitors like Aero Water Generators and Uravu Labs, Vayujal's adaptable machinery excels in footprint efficiency, requiring significantly less space to produce the same amount of water. This efficiency is particu-

larly valuable in urban areas where space is at a premium.

Majik Water, a Kenyan social enterprise, also uses innovative air-to-water technology to address water scarcity. They employ advanced dehumidification techniques to extract and condense moisture from the air, producing purified water. This process involves capturing humidity with industrial fans, condensing it through a refrigeration system, and filtering it using a reverse osmosis (RO) system to ensure purity. The water is also enriched with essential nutrients for consumption.

Njeri Nyoro, a representative of Majik Water, emphasises the company's commitment to long-term solutions for combating water scarcity: "Majik Water's unwavering dedication to sustainable water generation in an era of climate uncertainty serves as a testament to its mission."

Their systems are designed to be adaptable to various energy sources, including off-grid solar systems, making them suitable for regions with abundant sunshine. Majik Water offers a range of system sizes to cater to diverse needs, from individual households to commercial enterprises. With its technology, cost-effectiveness, and focus on sustainability, Majik Water provides a hopeful solution in a world facing water scarcity challenges.

Cost-effective solutions for the future

Imagine a world where water flows freely from the very air we breathe, eliminating the need for costly pipelines and extensive transportation networks. This groundbreaking water-from-air technology not only

18%

of the world Population

4%

of the world's
freshwater resources

40%

Indian with no access
to clean water by 2030

70%

of India's water is
contaminated

Water in India (Credit: Composite Water Management Index, Niti Aayog, August 2019)